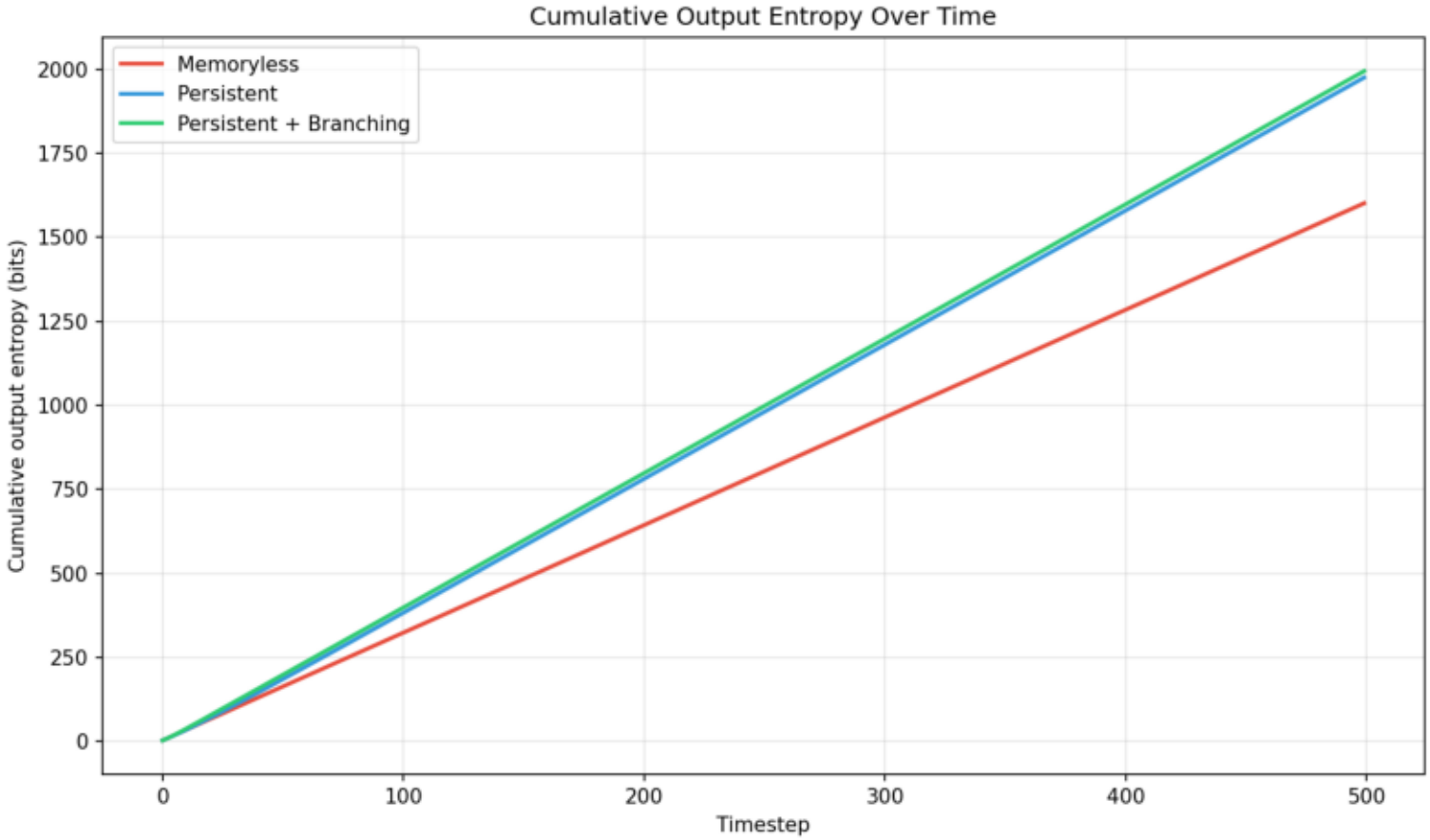
Entropy Flow Model

Complete Plot Collection

Layer 1: Base Model (13 plots)
Layer 2: Evolution Extension (7 plots)
Layer 3: Paper Experiment Suite (8 plots)

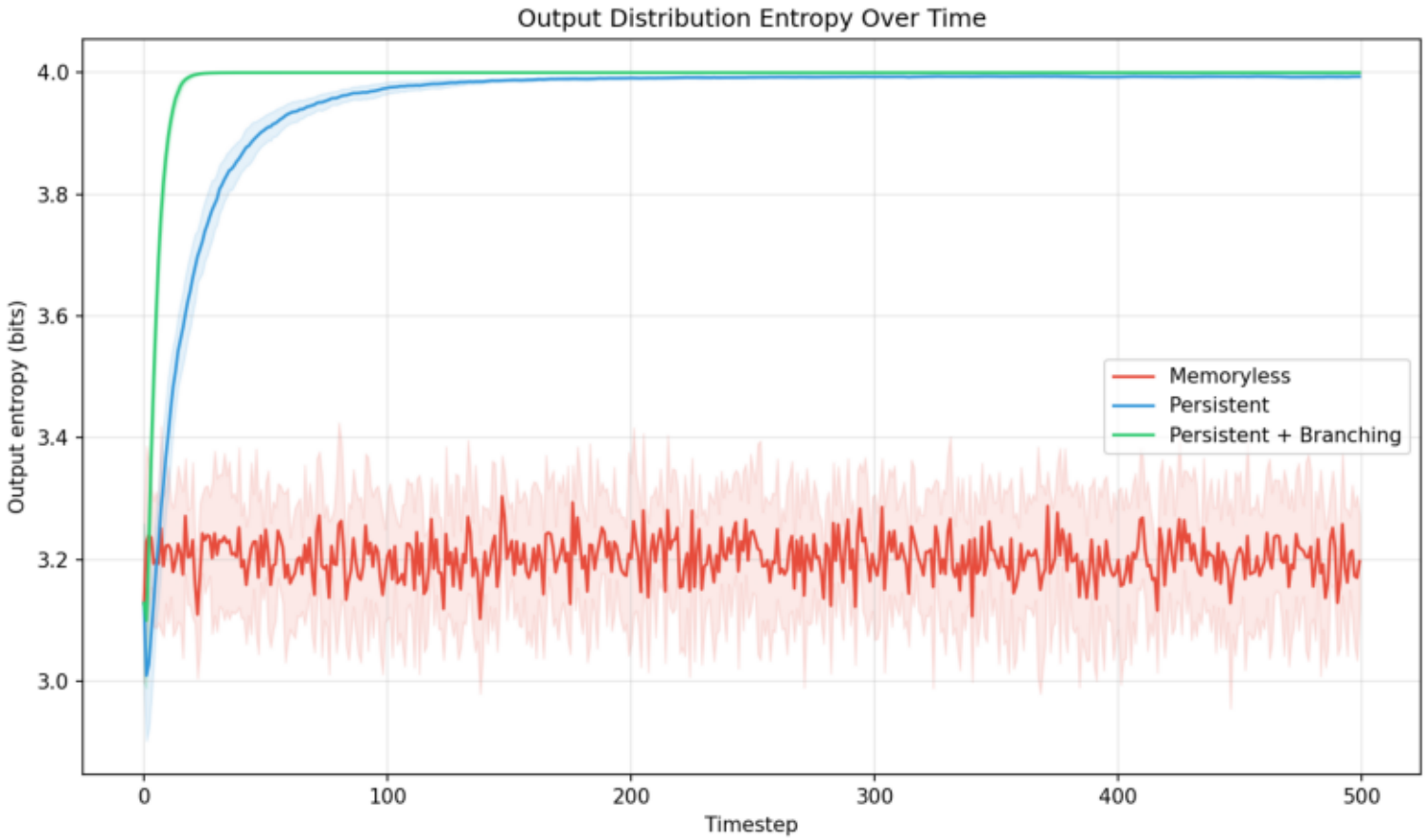
28 figures total

Layer 1: Base Model



Cumulative output entropy across three regimes

plots/cumulative_entropy.png



Instantaneous output entropy per step

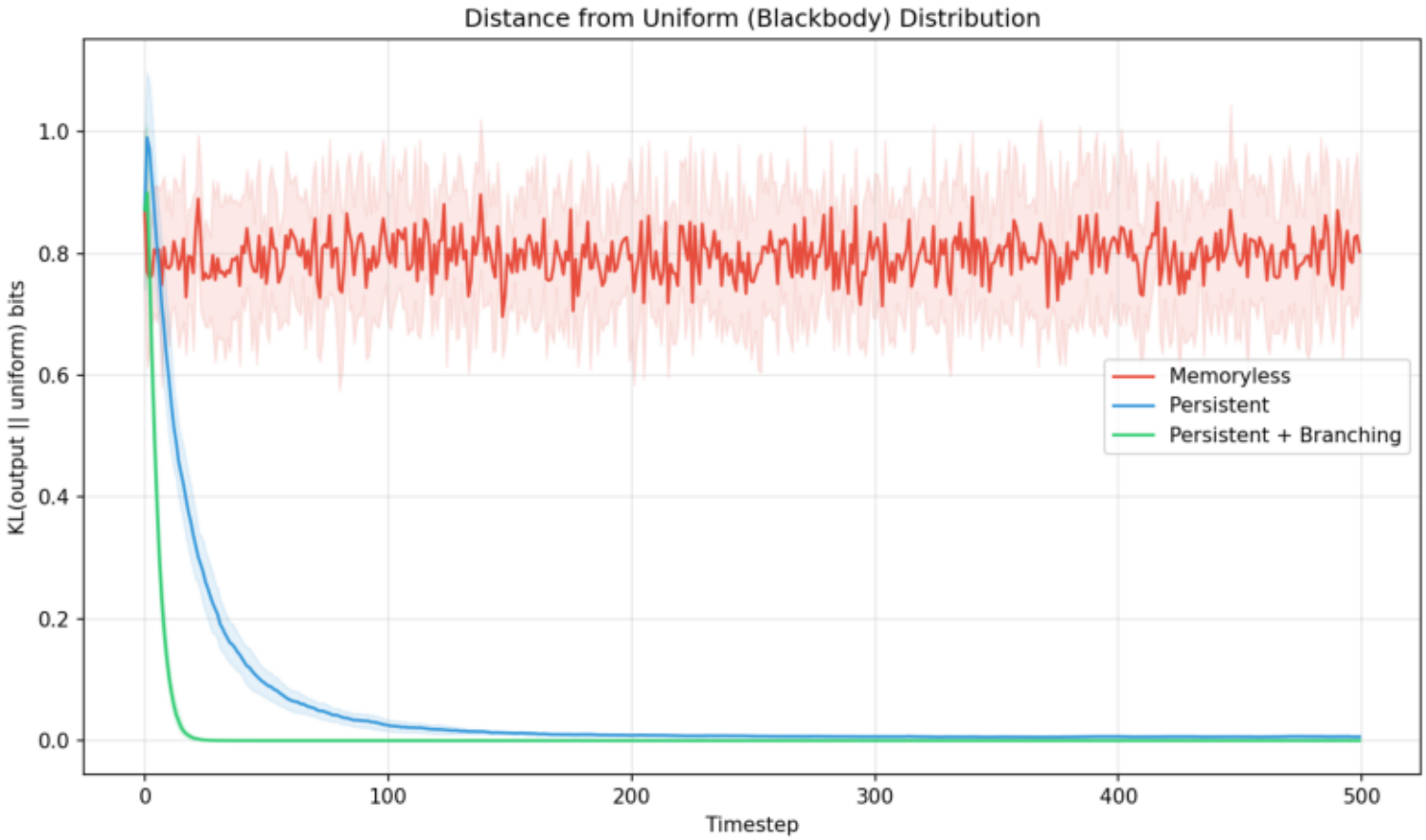
plots/output_entropy.png

Output fragmentation over time

plots/fragmentation.png

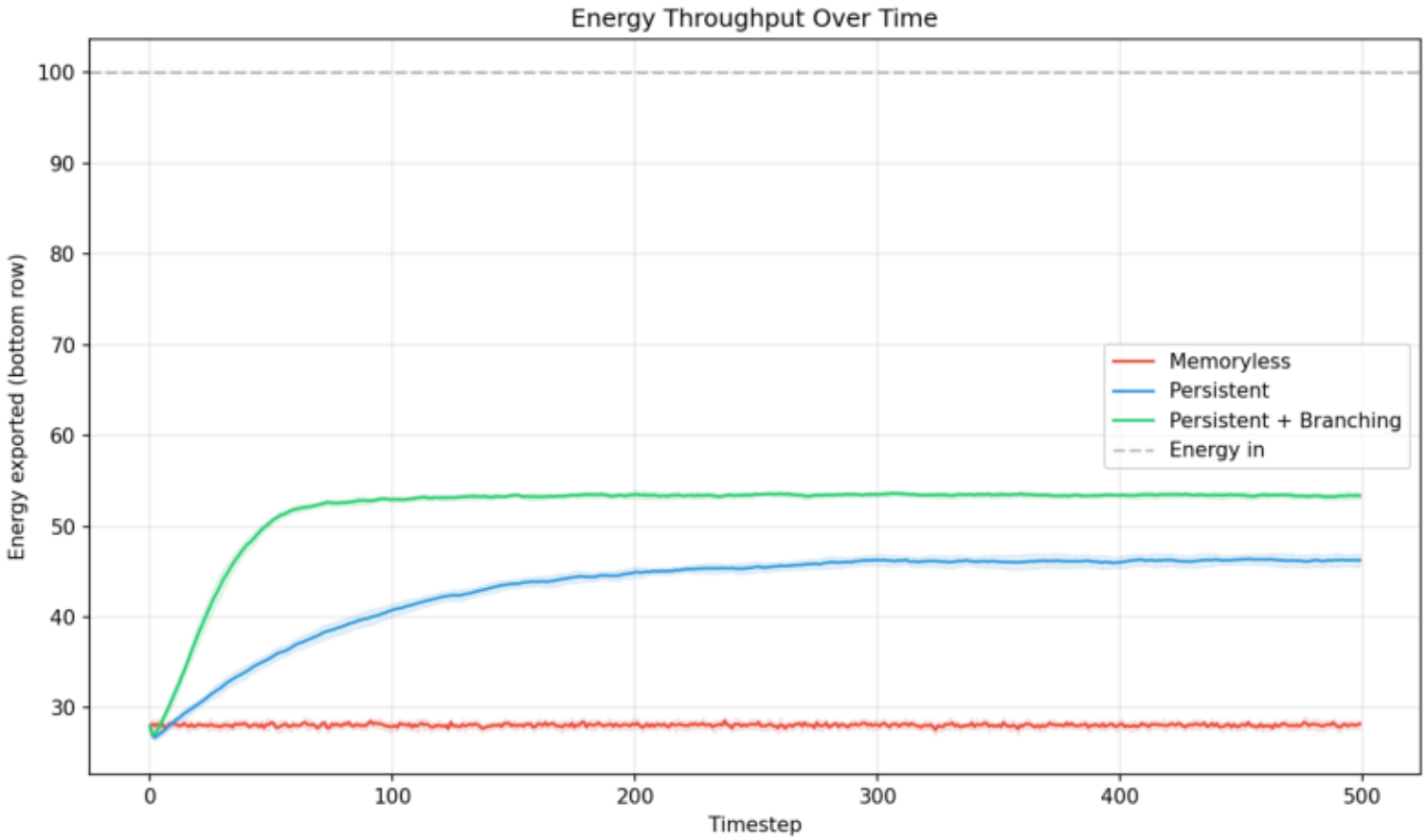
Effective number of output modes

plots/effective_modes.png



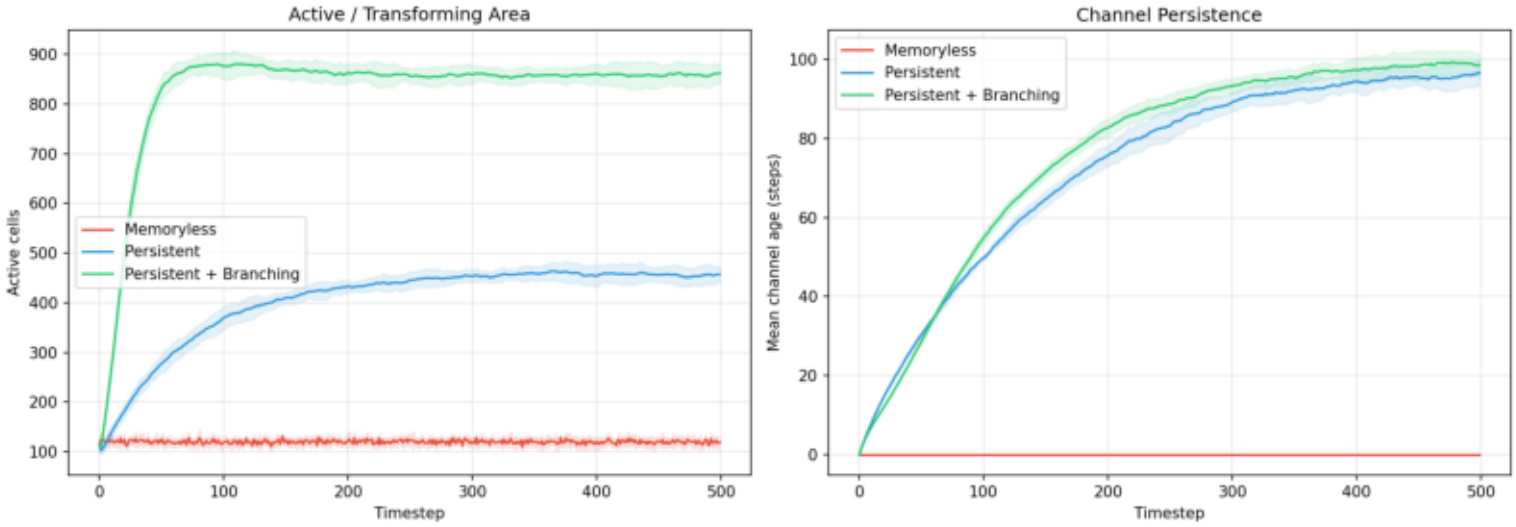
KL divergence from uniform distribution

plots/kl_divergence.png



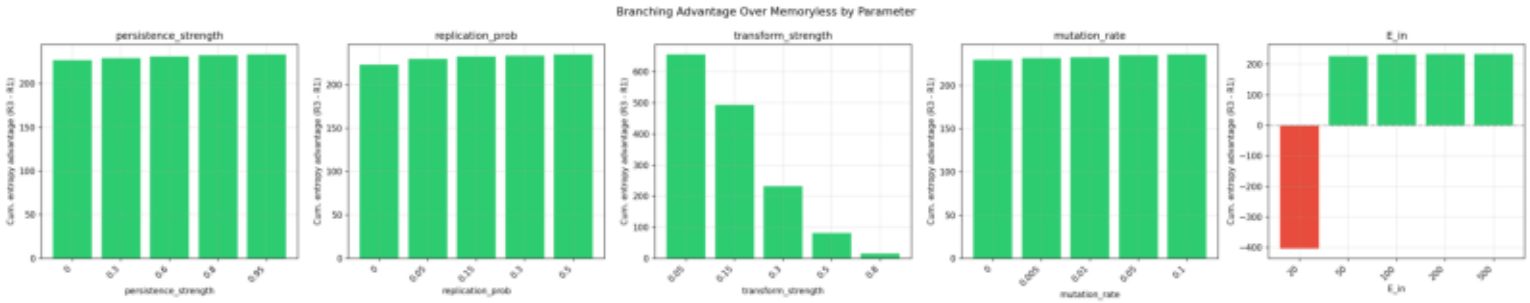
Energy throughput and balance

plots/energy_balance.png



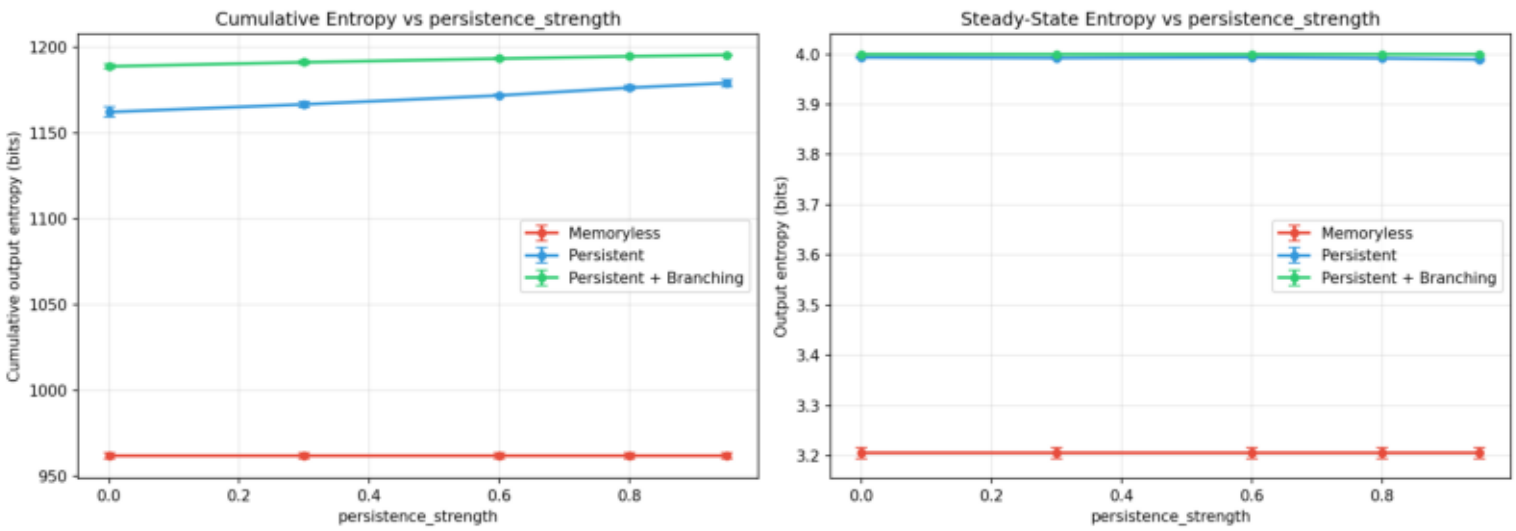
Active area and channel age

plots/structure_persistence.png



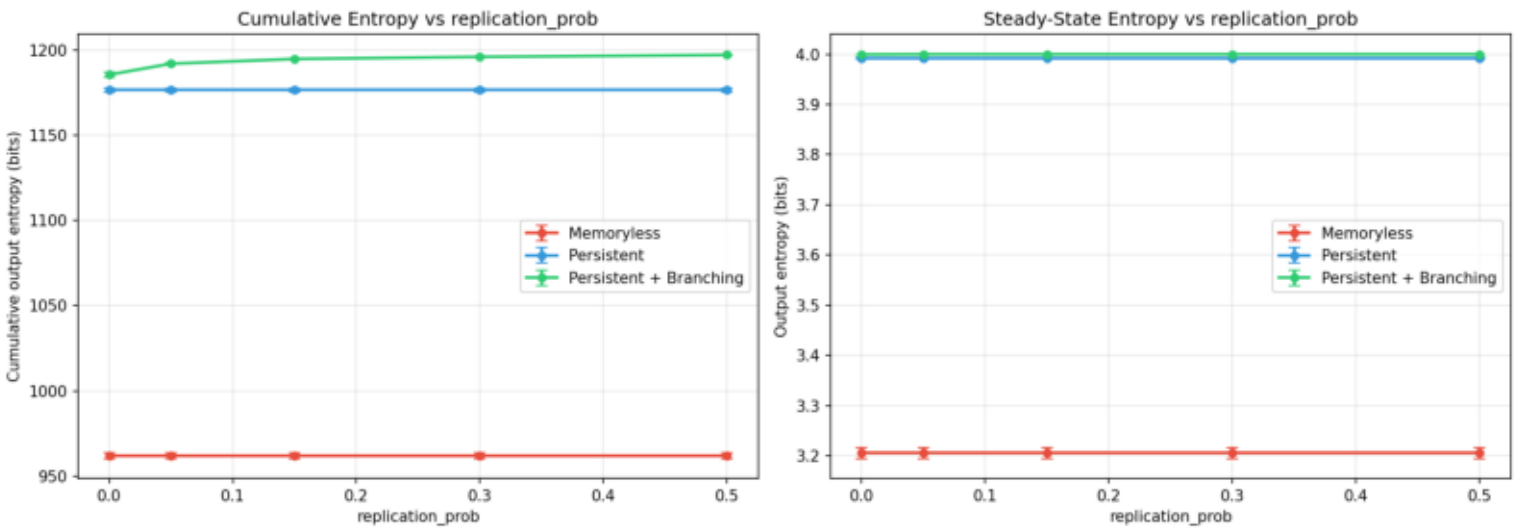
Branching advantage across parameters

plots/phase_diagram.png



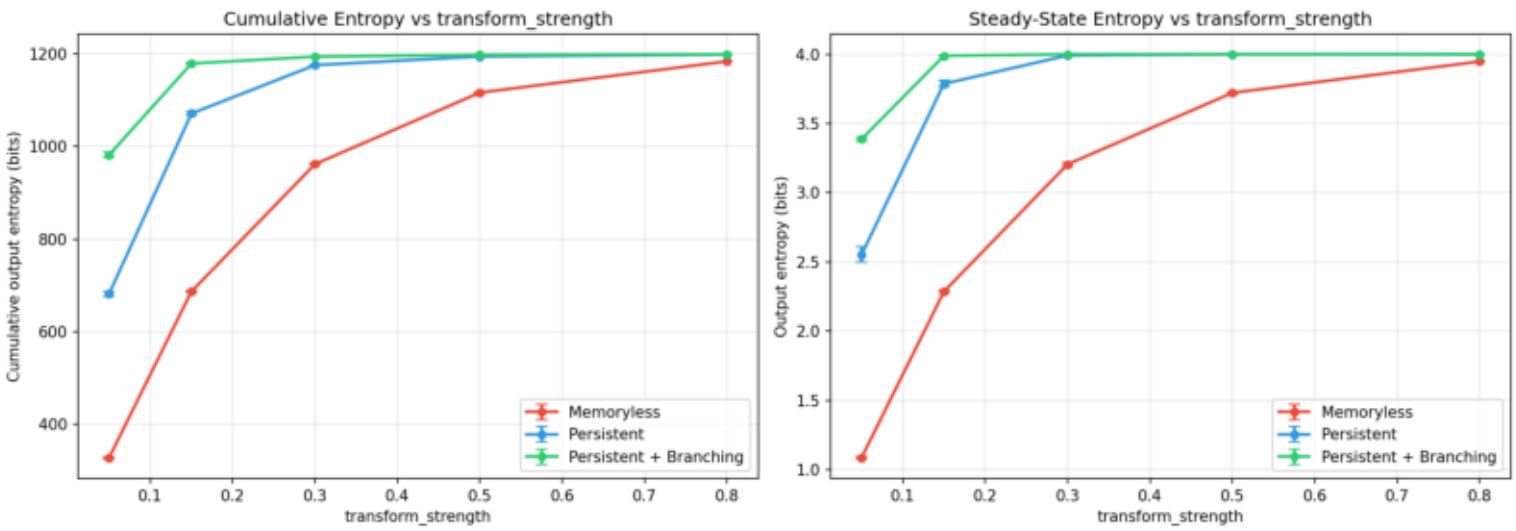
Parameter sweep: persistence strength

plots/sweep_persistence_strength.png



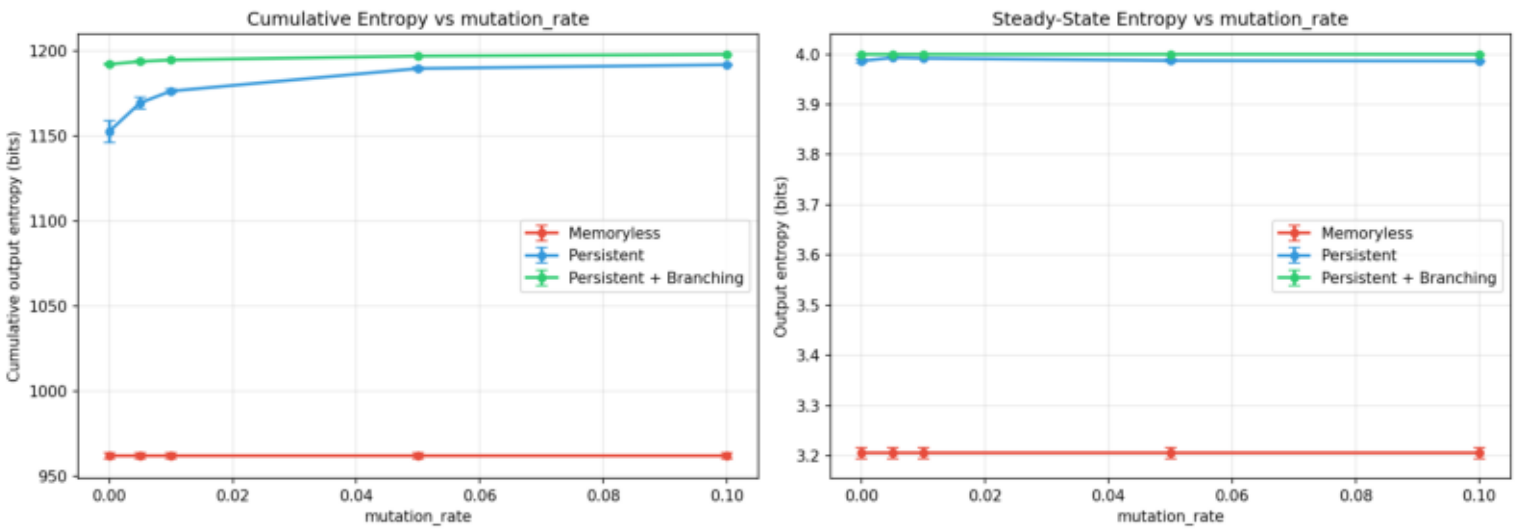
Parameter sweep: replication probability

plots/sweep_replication_prob.png



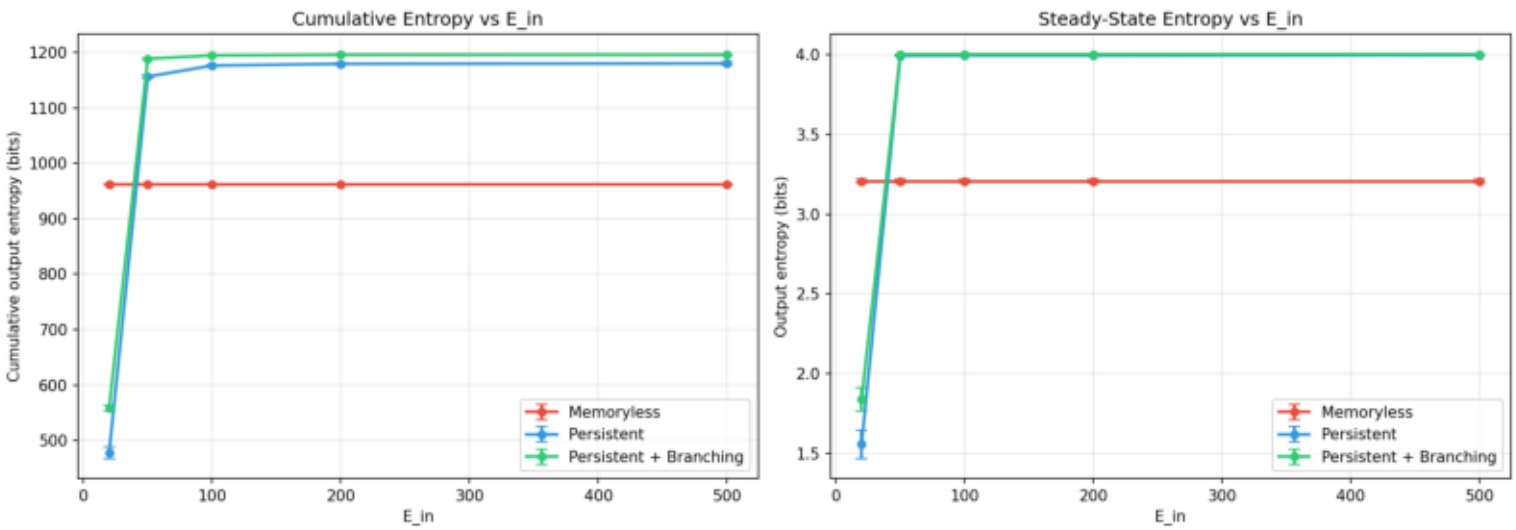
Parameter sweep: transform strength

plots/sweep_transform_strength.png



Parameter sweep: mutation rate

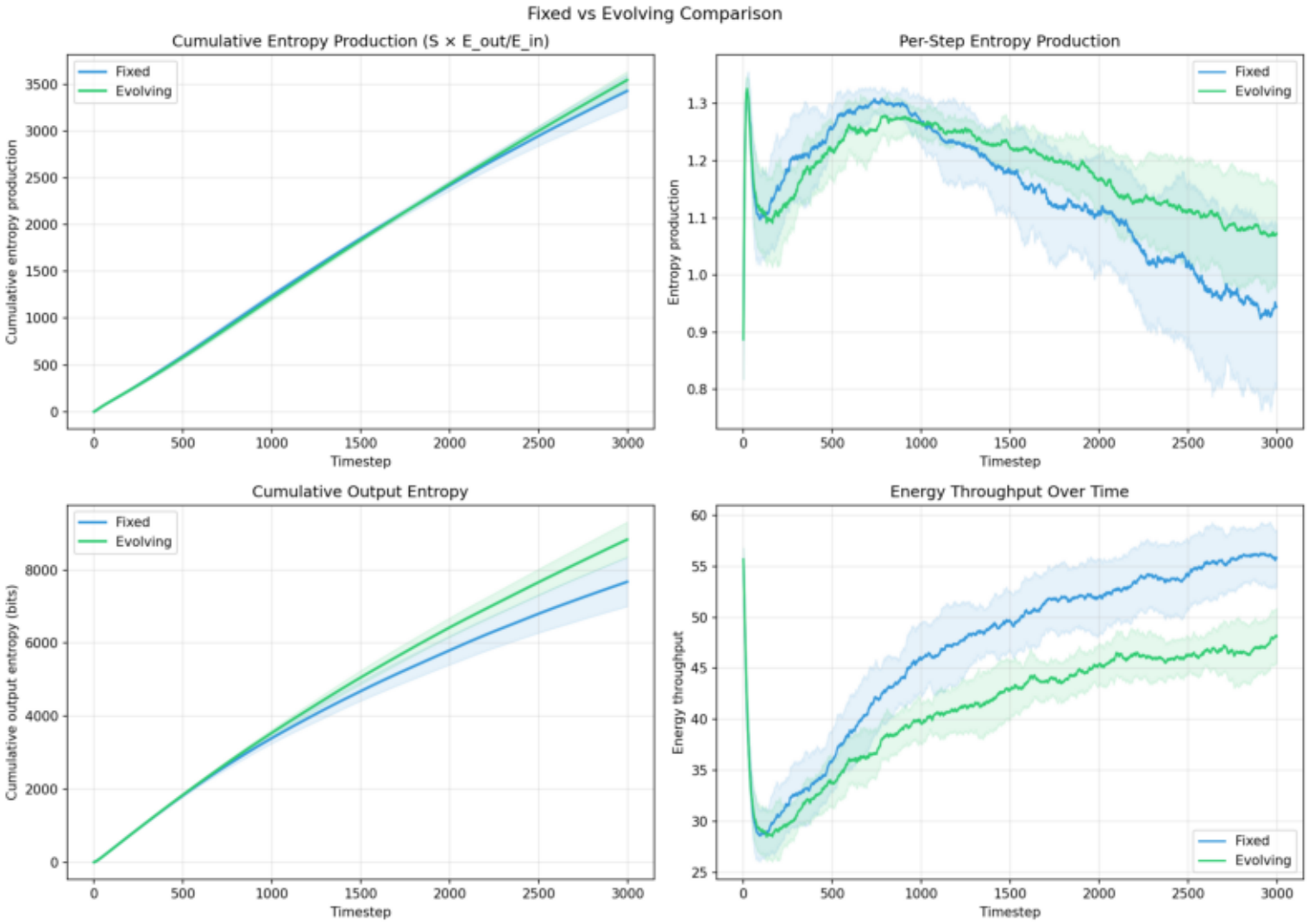
plots/sweep_mutation_rate.png



Parameter sweep: energy input

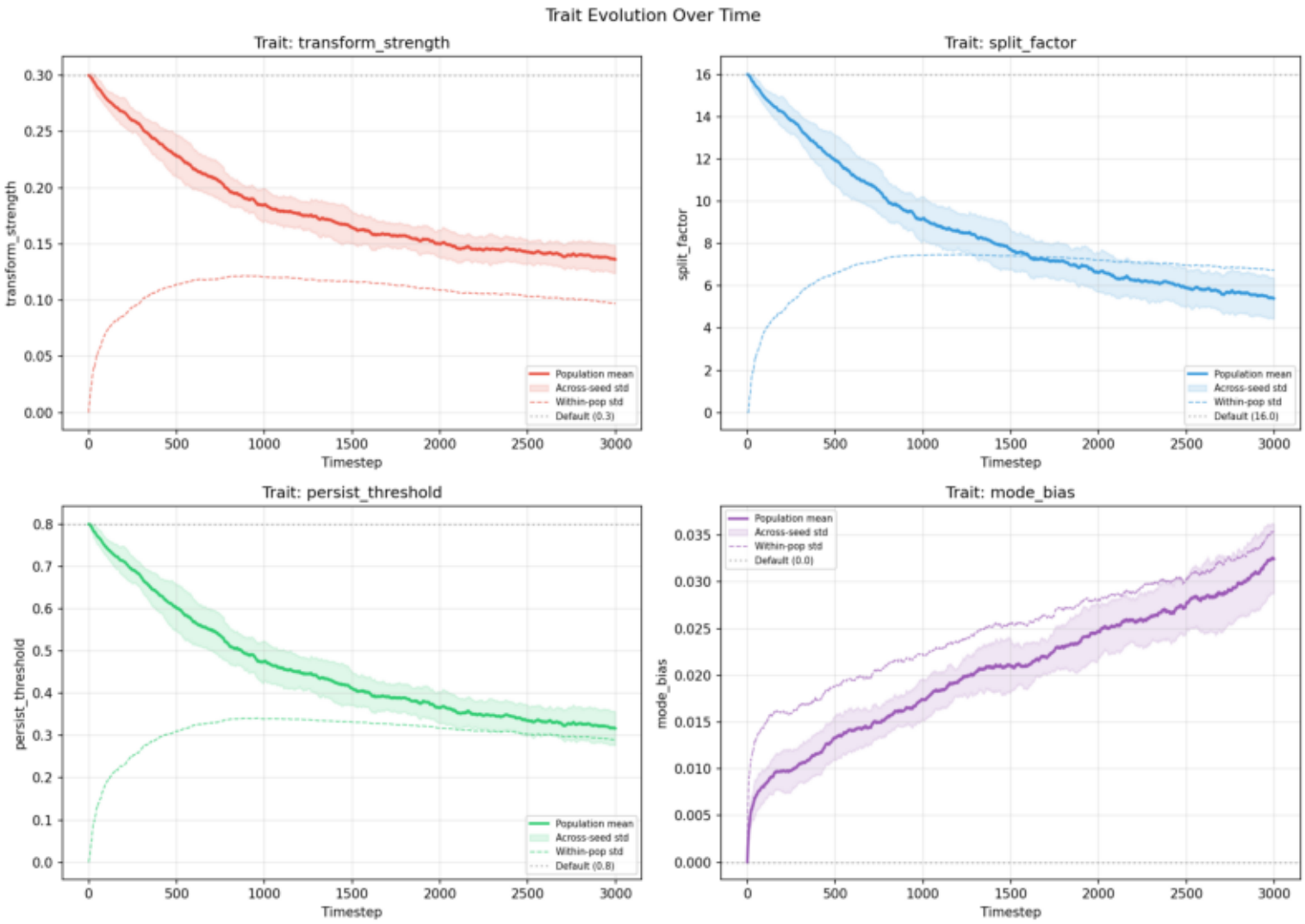
plots/sweep_E_in.png

Layer 2: Evolution Extension



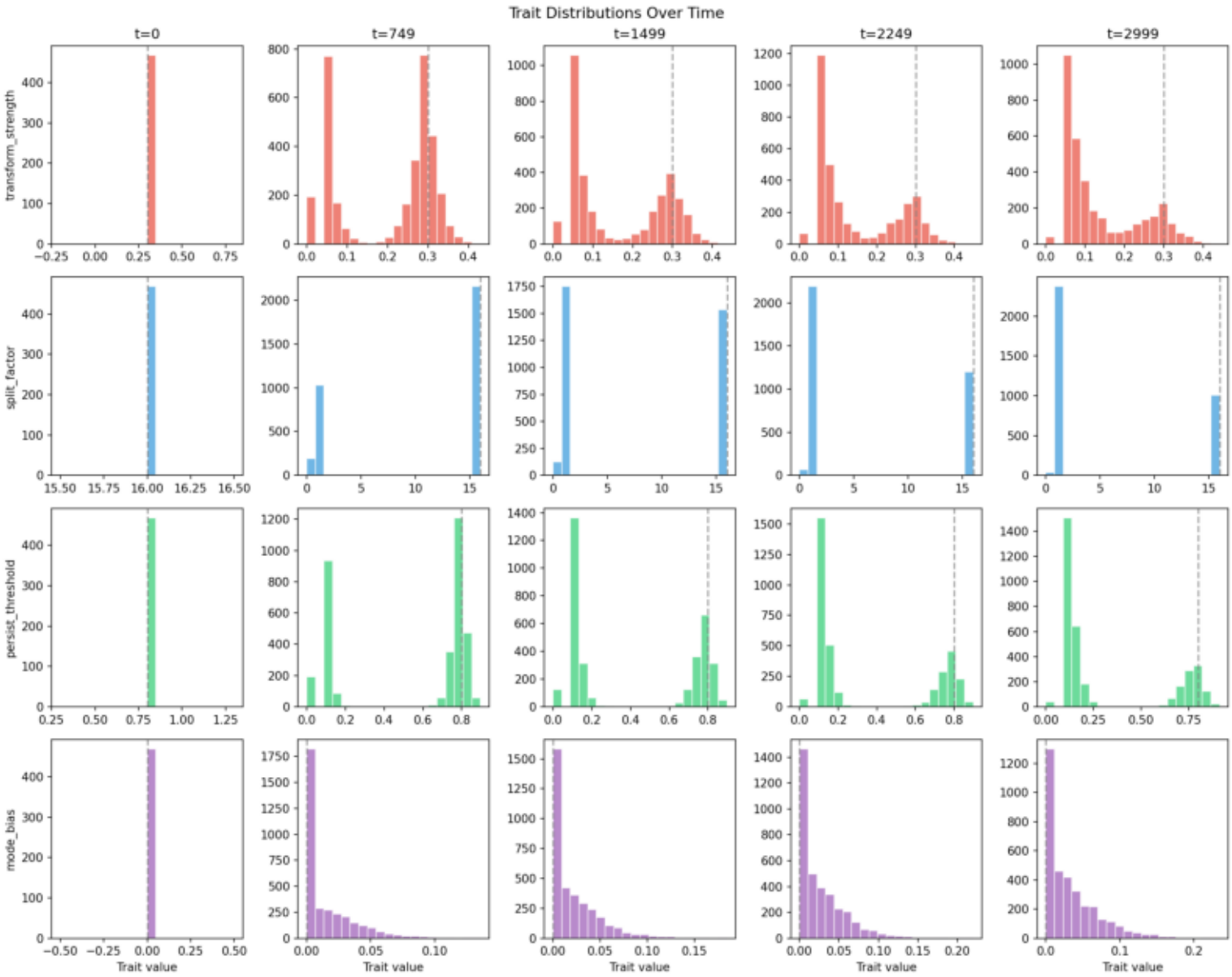
Fixed vs evolving: EP, entropy, throughput

evo_plots/fixed_vs_evolving.png



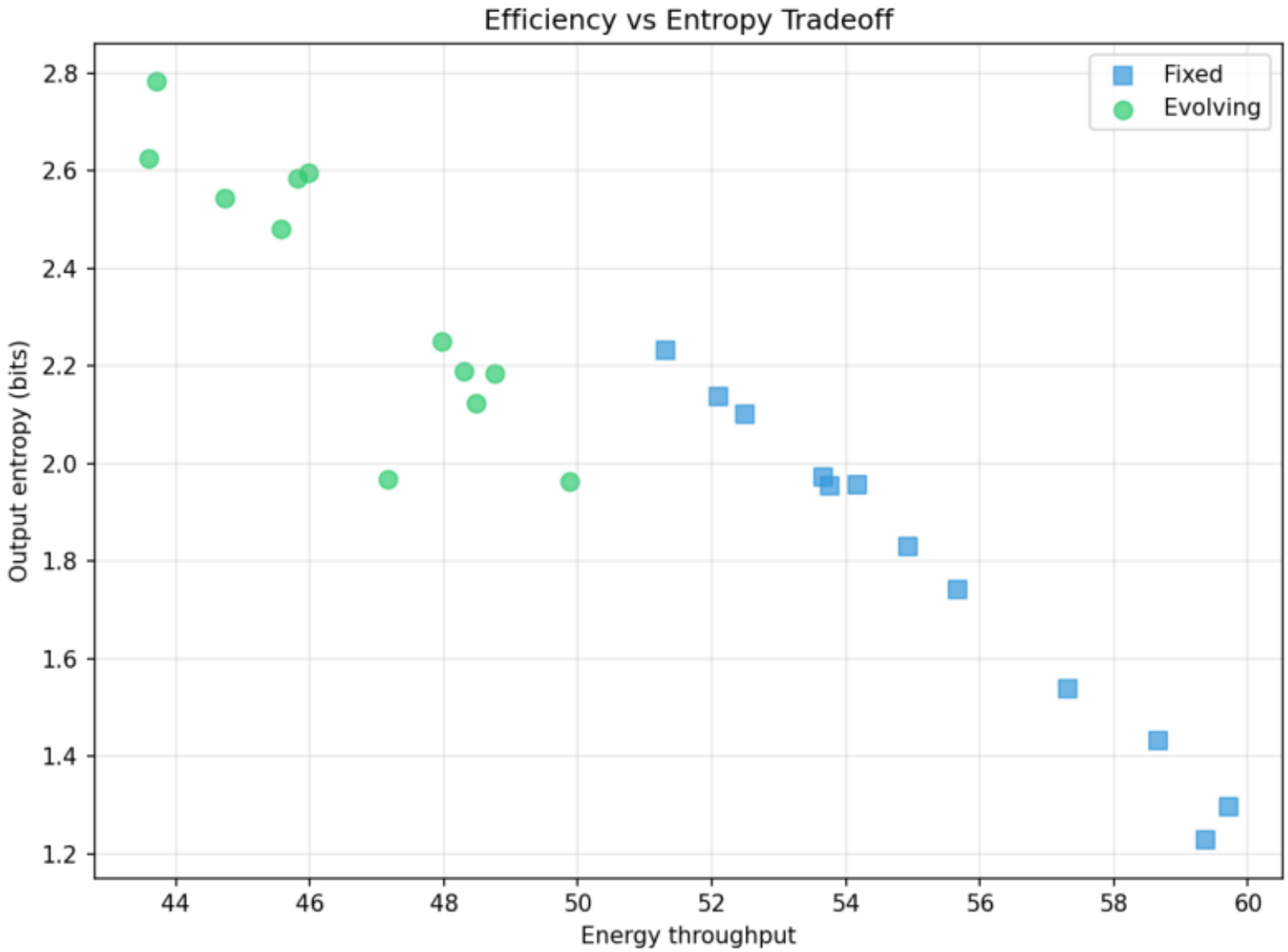
Trait evolution over time

evo_plots/trait_evolution.png



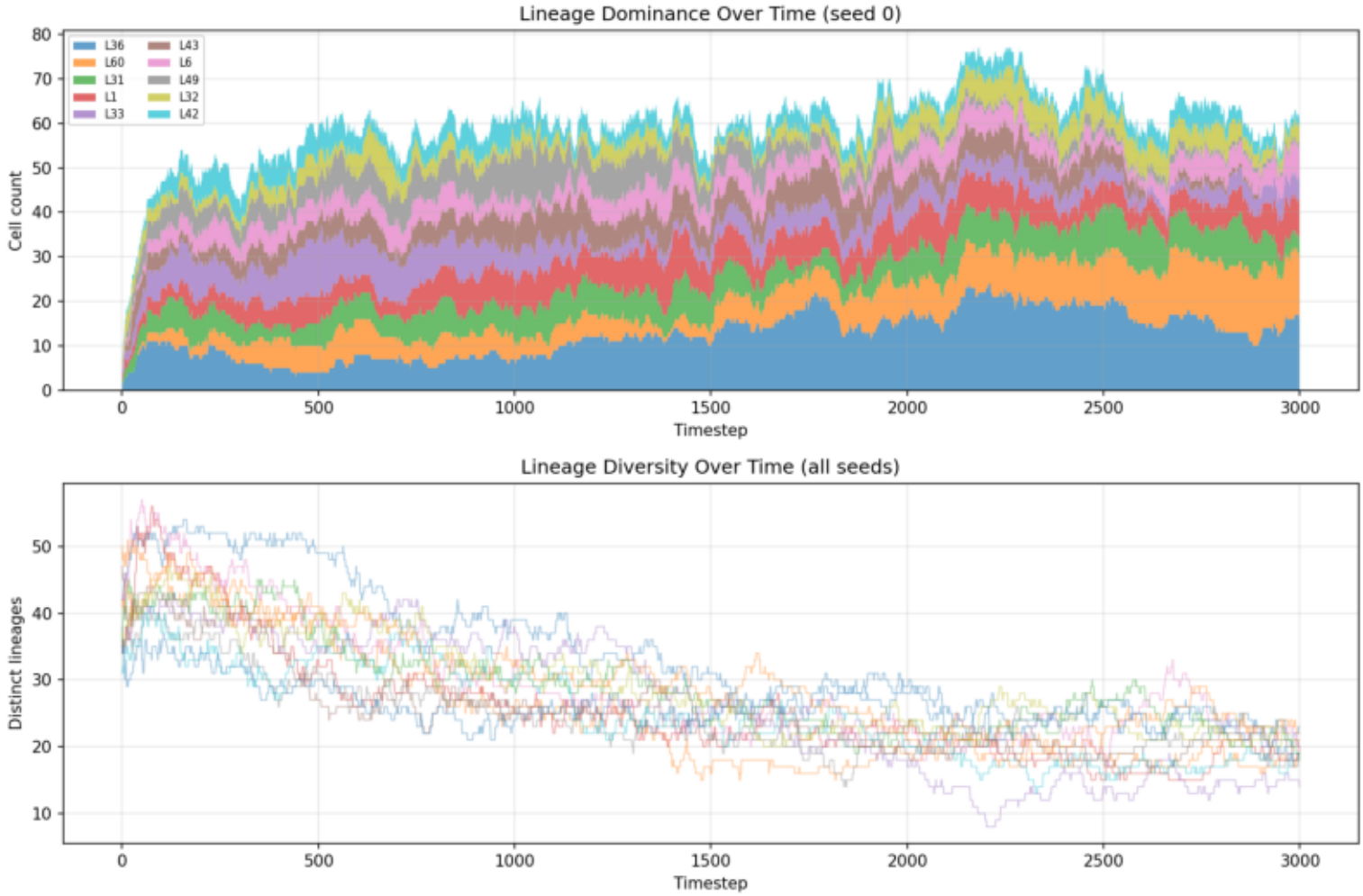
Trait distributions

evo_plots/trait_histograms.png



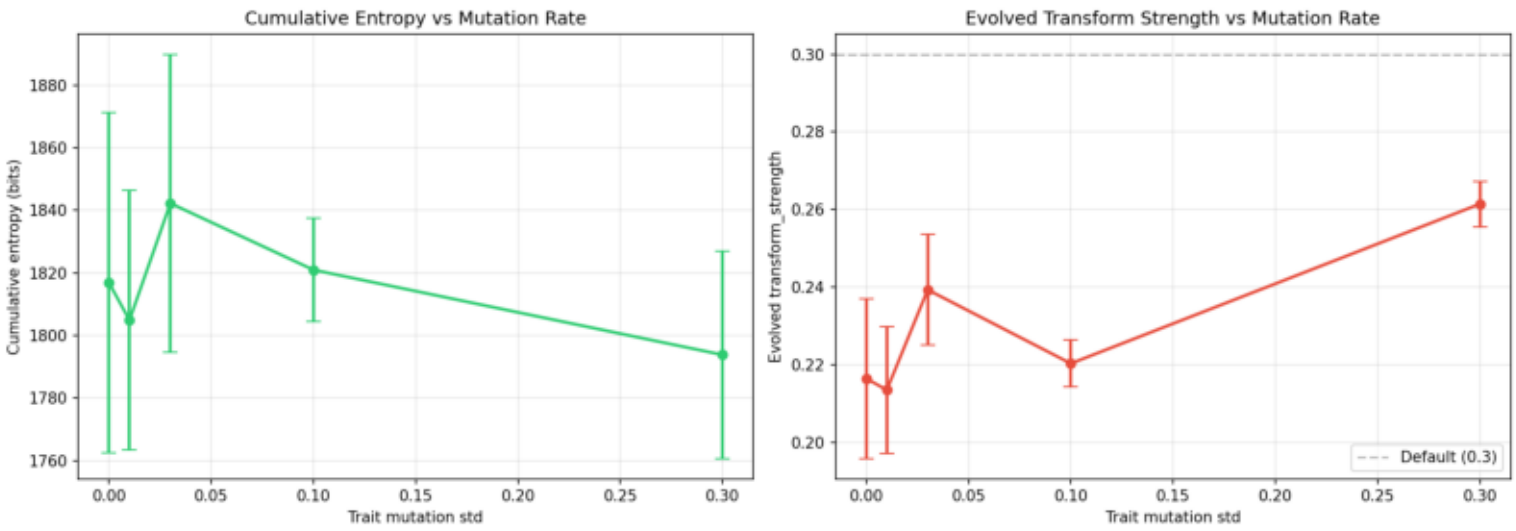
Energy-entropy tradeoff

evo_plots/energy_entropy_tradeoff.png



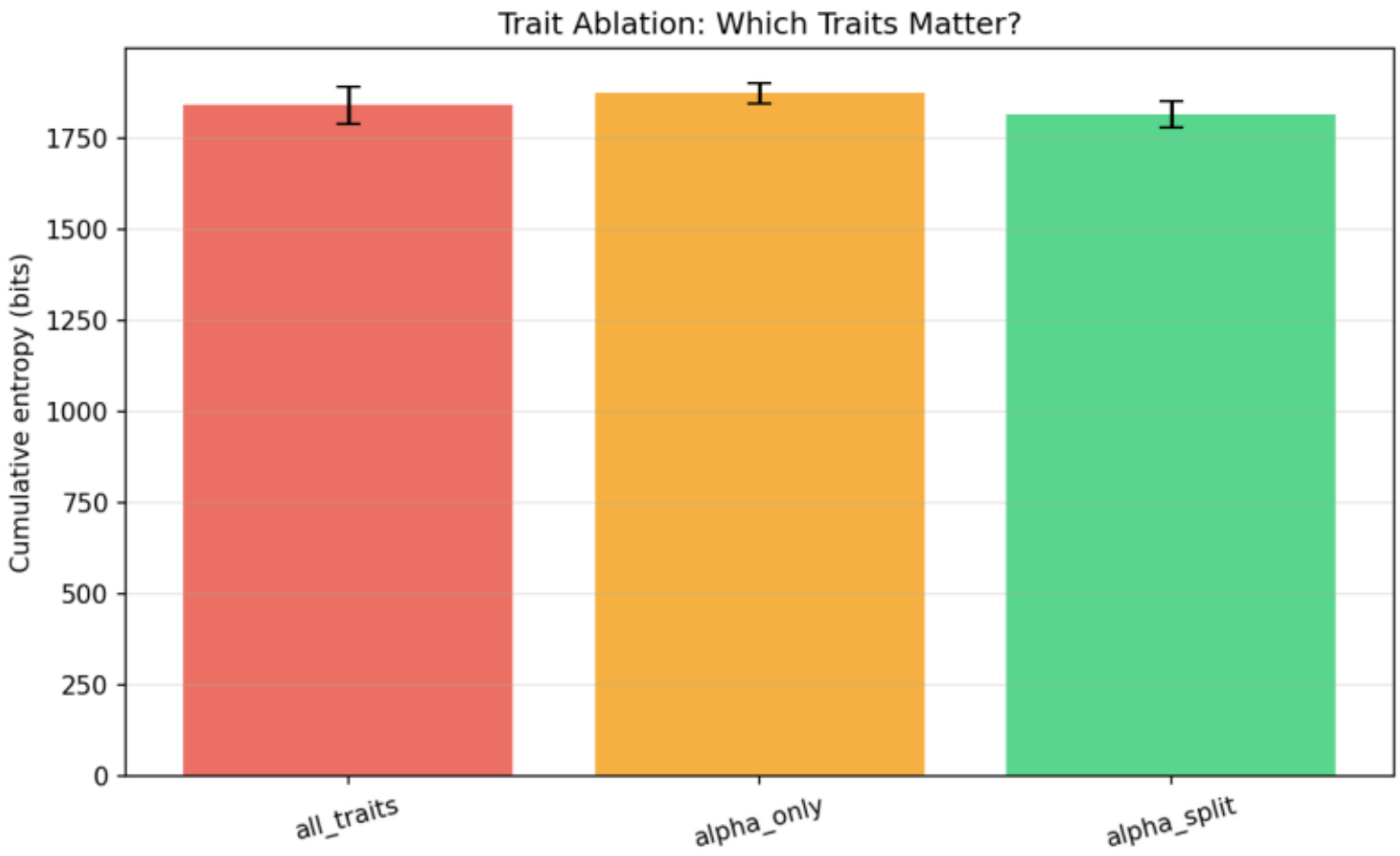
Lineage diversity over time

evo_plots/lineage_dominance.png



Mutation rate sweep

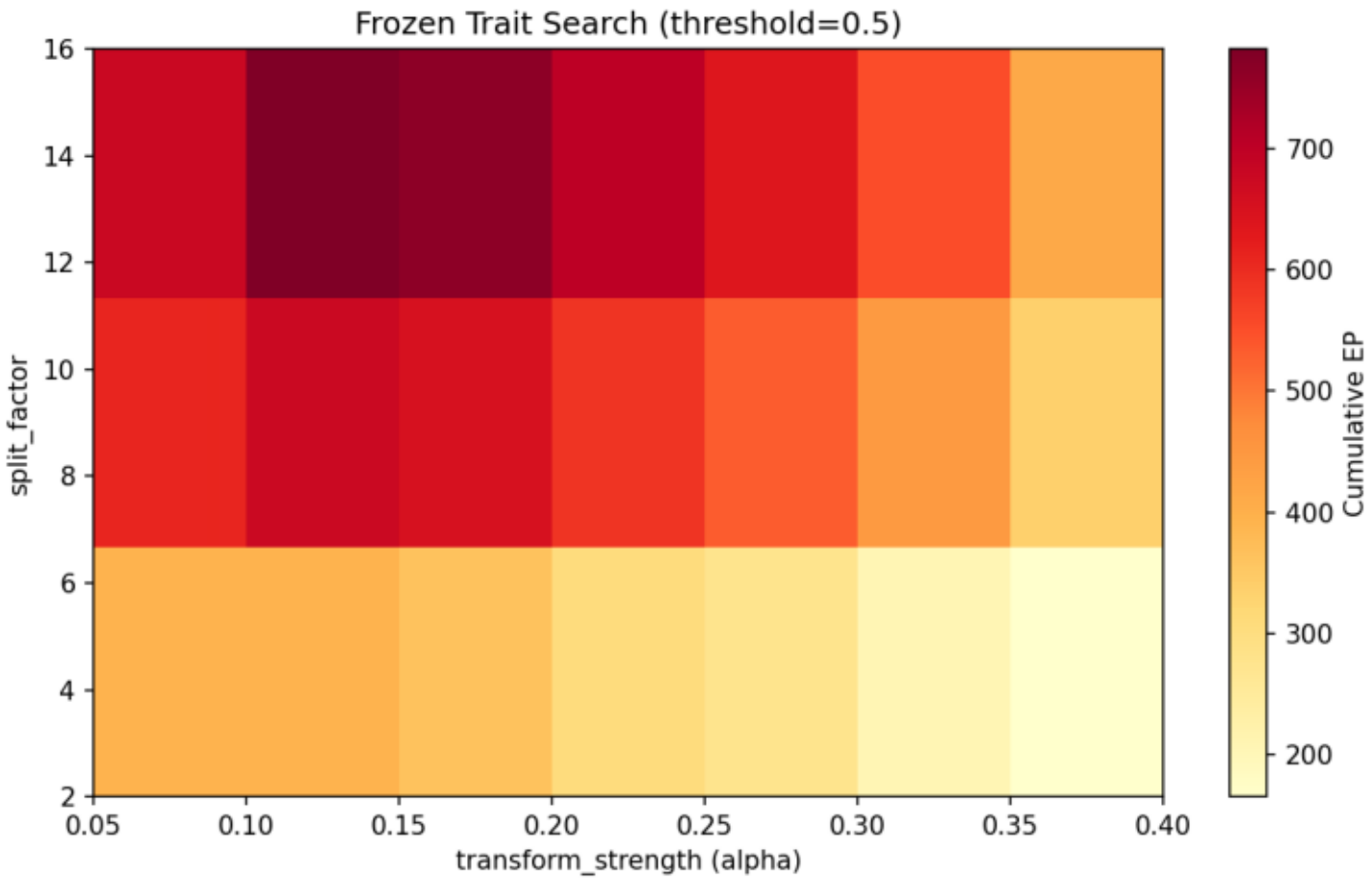
evo_plots/mutation_sweep.png



Trait ablation results

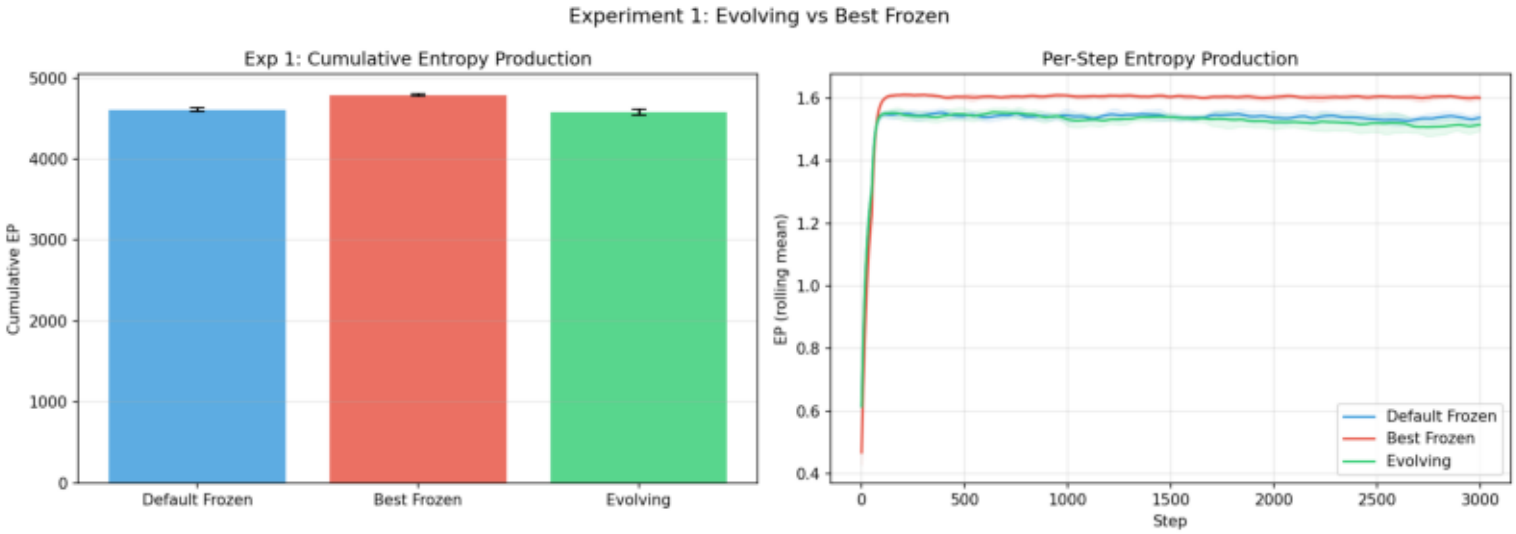
evo_plots/trait_ablation.png

Layer 3: Paper Experiment Suite



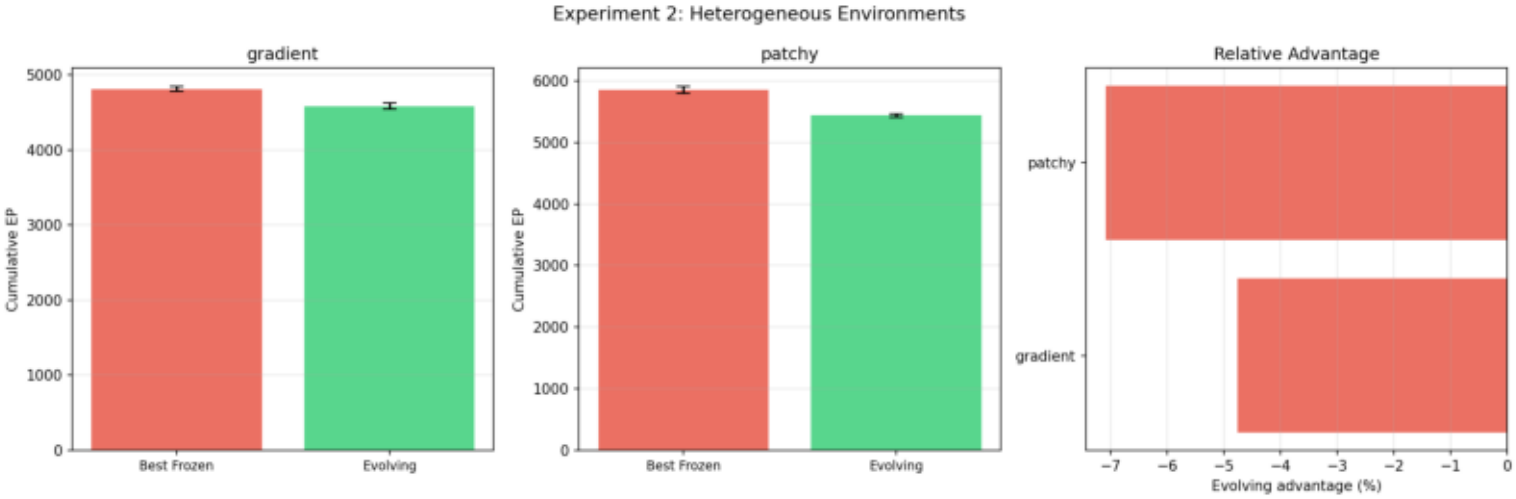
Exp 1: Frozen trait landscape (63 combinations)

suite_plots/exp1_landscape.png



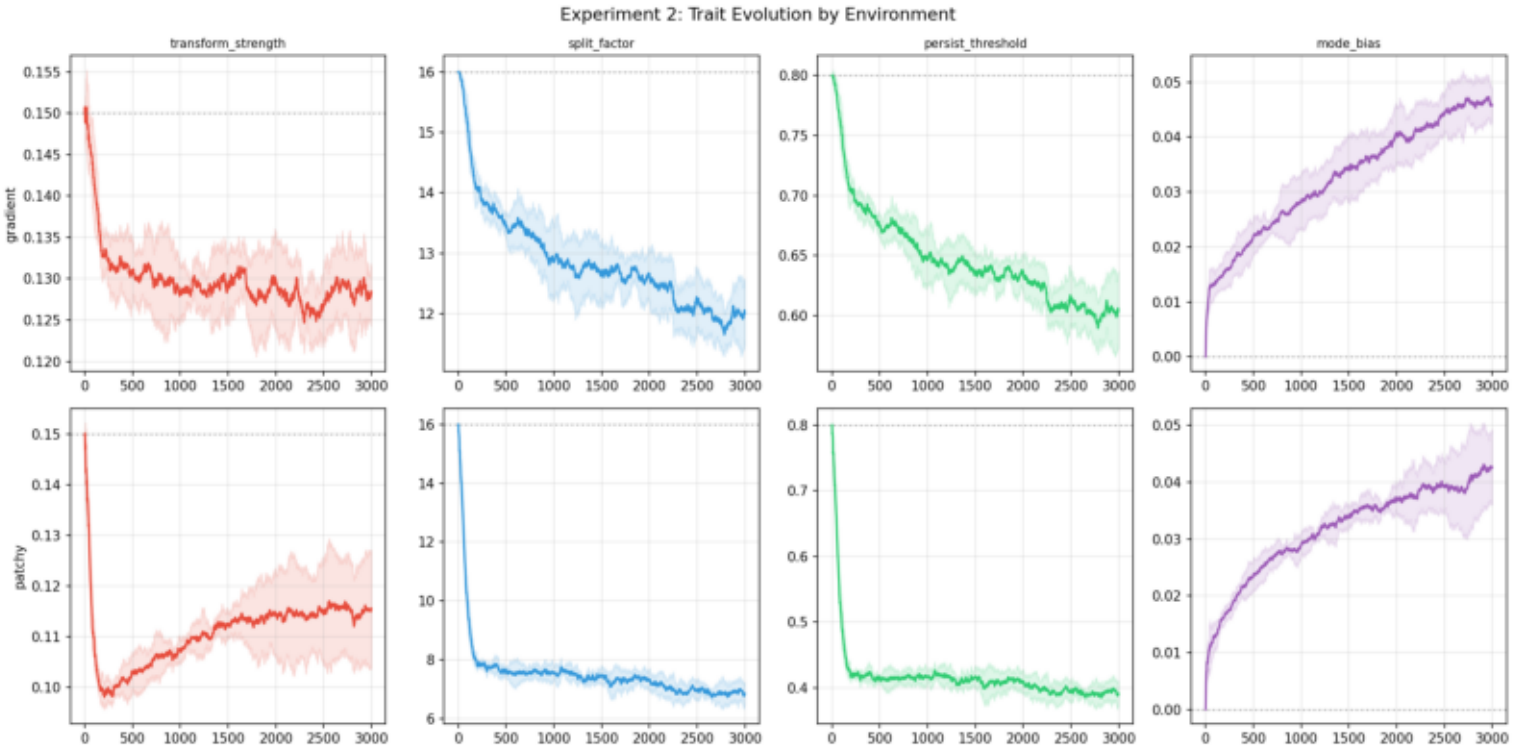
Exp 1: Evolving vs frozen baselines

suite_plots/exp1_comparison.png



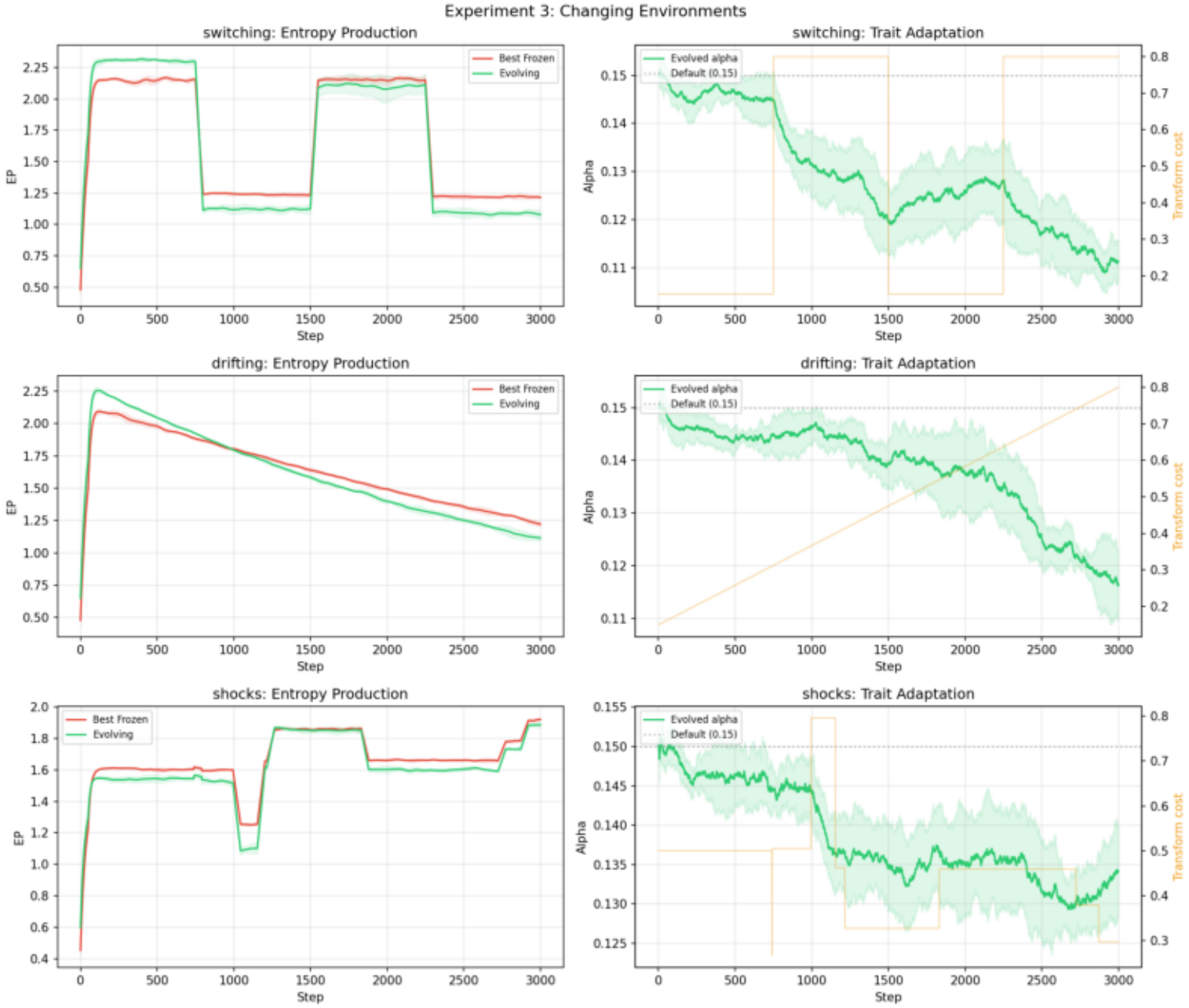
Exp 2: Heterogeneous environments

suite_plots/exp2_environments.png



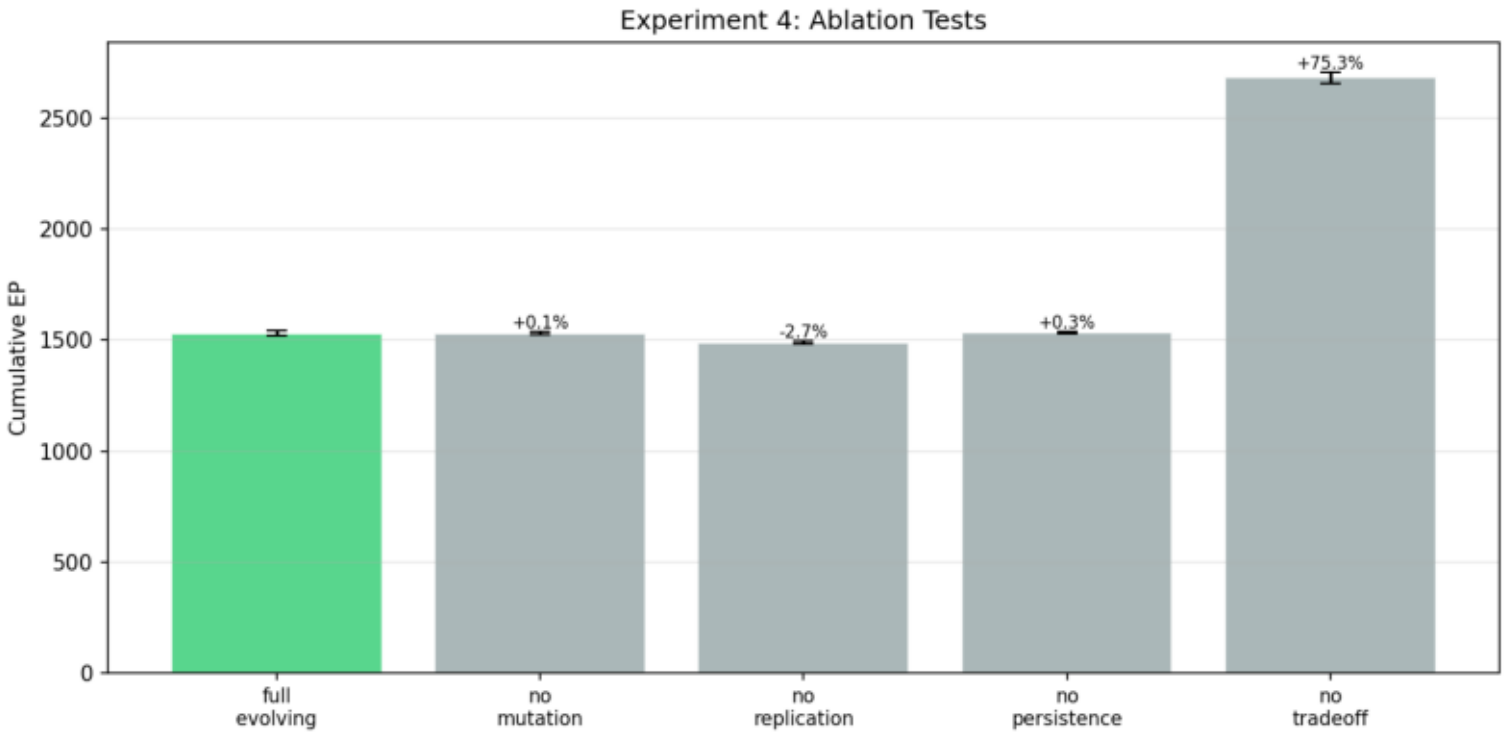
Exp 2: Trait evolution by environment

suite_plots/exp2_traits.png



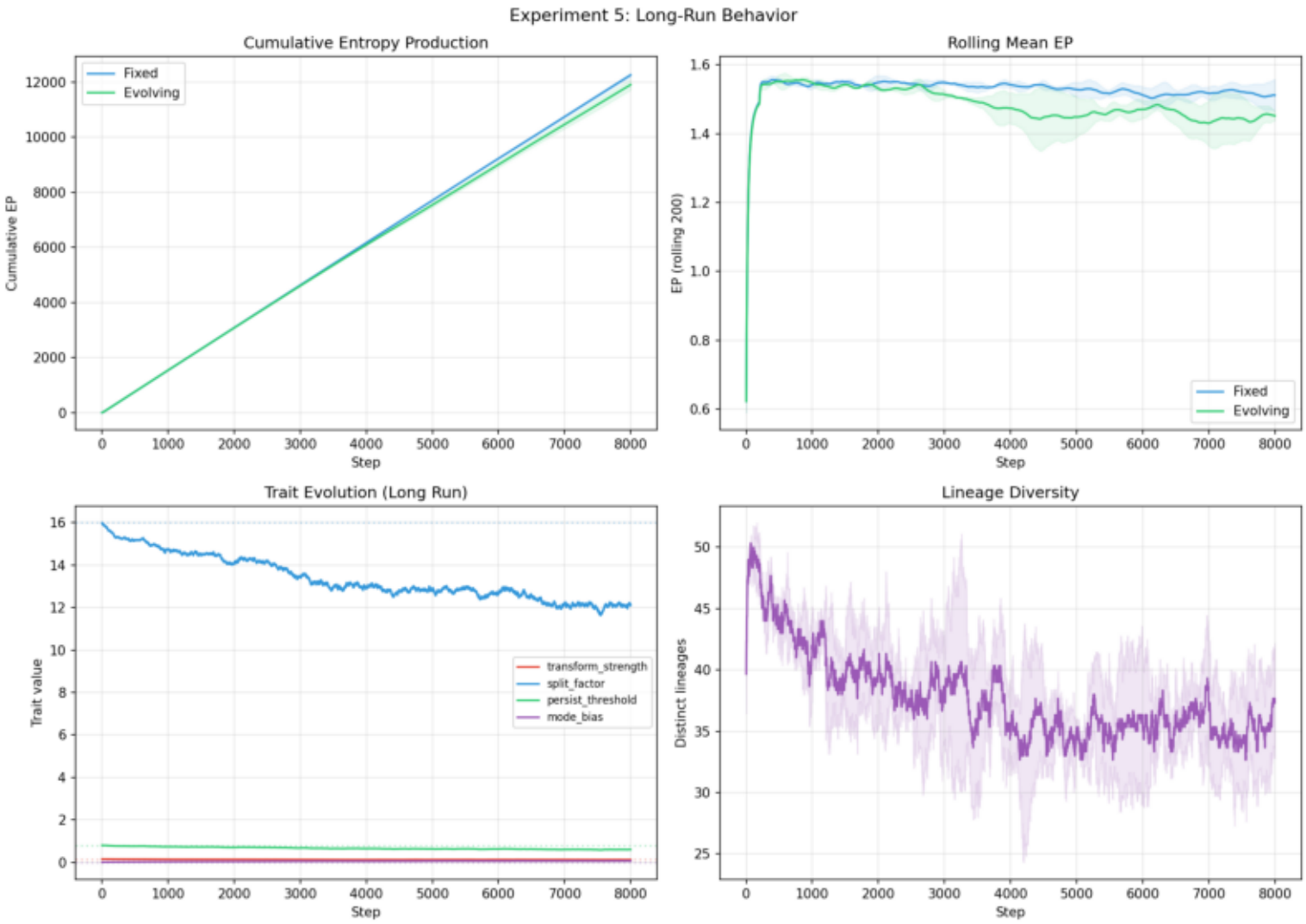
Exp 3: Changing environments time series

suite_plots/exp3_timeseries.png



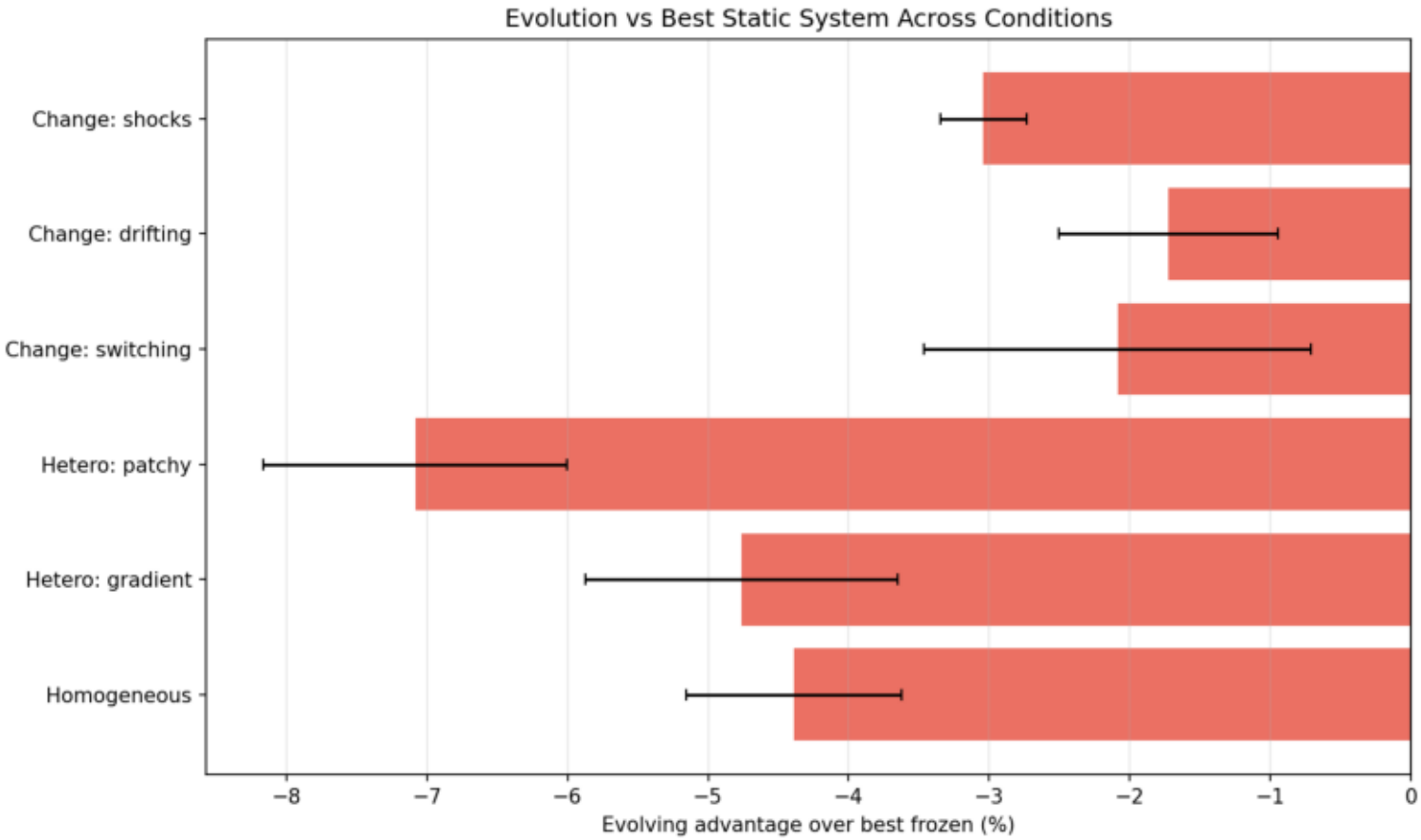
Exp 4: Ablation tests

suite_plots/exp4_ablation.png



Exp 5: Long-run behavior (8000 steps)

suite_plots/exp5_longrun.png



Summary: Evolving advantage across all conditions

suite_plots/summary_figure.png